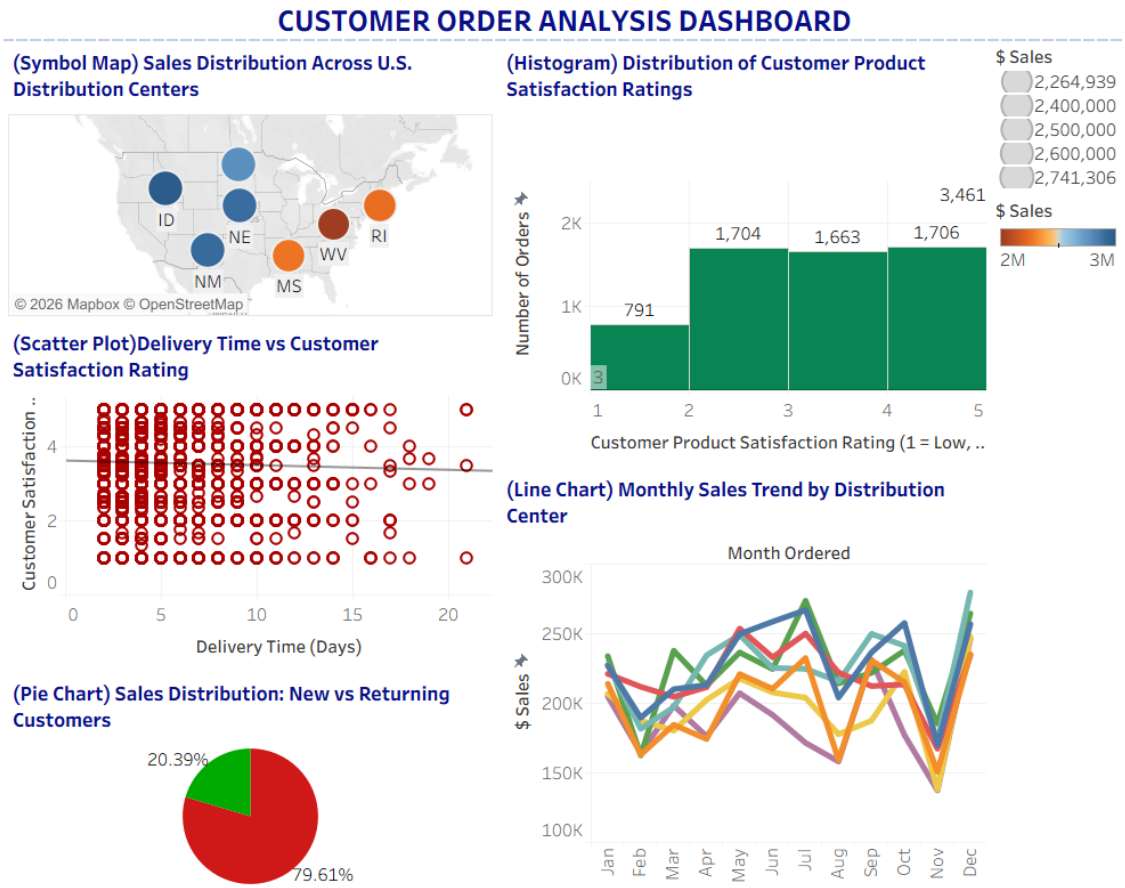
WEEK 2 ASSIGNMENT
TABLEAU DATA VISUALIZATION REPORT
RABLEEN KAUR KAPOOR



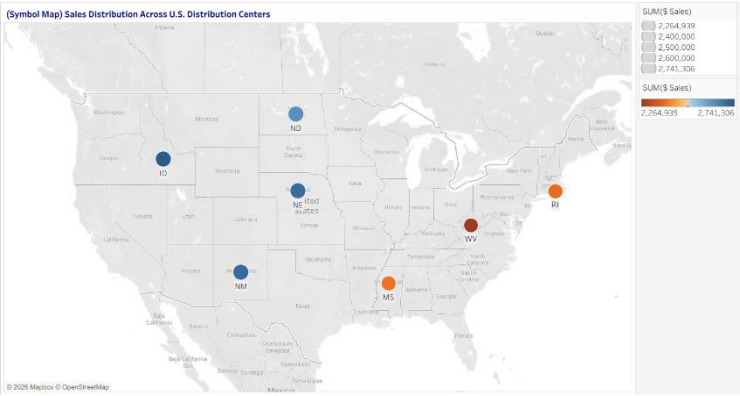
[Tableau Dashboard Public Link <click here>]

This dataset represents sales transactions, customer behavior, and operational performance related to product distribution across multiple distribution centers in the United States. The data includes information such as sales amounts, order dates, delivery times, distribution center locations, customer types, and customer satisfaction ratings. By analyzing these variables, the dataset provides insights into how sales performance, customer experience, and distribution operations interact within the business.

The objective of this Tableau analysis is to transform the raw Excel data into meaningful visual insights that help explain patterns and relationships within the dataset. Using Tableau, several visualizations were created to explore geographic sales distribution, monthly sales trends, customer segmentation, satisfaction levels, and the relationship between delivery time and customer satisfaction. These visual tools allow complex data to be interpreted more easily and support better understanding of operational and customer-related patterns.

(Symbol Map) Sales Distribution Across U.S. Distribution Centers

[Tableau Public Link <click here>]

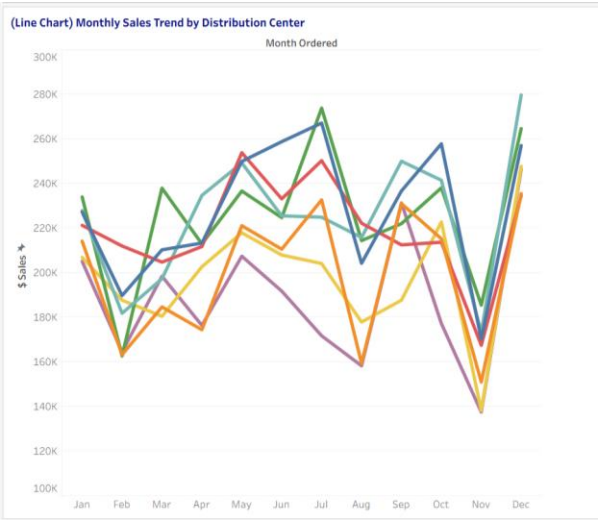


- The symbol map visualizes how sales are distributed across different distribution centers located throughout the United States. Each location marker represents a distribution center, allowing viewers to compare regional sales performance geographically.
- The map highlights the spatial differences in sales activity, showing which regions contribute more strongly to overall sales. Larger or more prominent markers indicate areas with higher sales volume.
- This visualization helps identify regional demand patterns and operational coverage, enabling businesses to understand where their products are performing best geographically.

Key Insight: Certain distribution centers generate noticeably higher sales, indicating that regional demand and distribution efficiency play an important role in overall business performance.

(Line Chart) Monthly Sales Trend by Distribution Center

[Tableau Public Link <click here>]

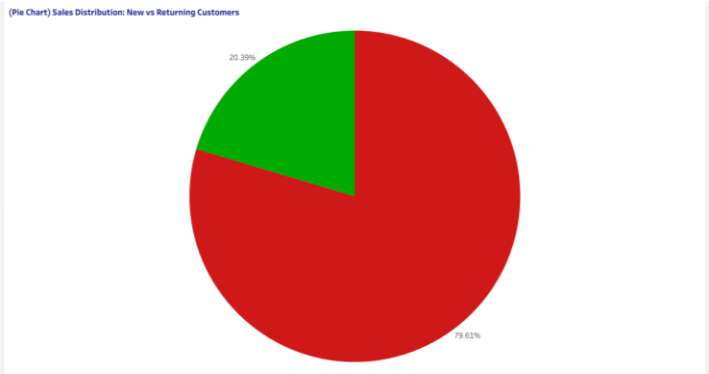


- The line chart displays how sales change over time across different distribution centers on a monthly basis. Each line represents a specific distribution center, allowing comparison of sales trends throughout the year.
- The visualization helps identify seasonal fluctuations and periods of higher or lower sales activity. Peaks and dips in the lines illustrate how demand changes across months.
- By comparing multiple lines simultaneously, the chart reveals differences in performance between distribution centers and how consistently each location contributes to sales.

Key Insight: Sales show noticeable monthly variations, suggesting that customer demand changes over time, and some distribution centers maintain more stable performance than others.

(Pie Chart) Sales Distribution: New vs Returning Customers

[Tableau Public Link <click here>]

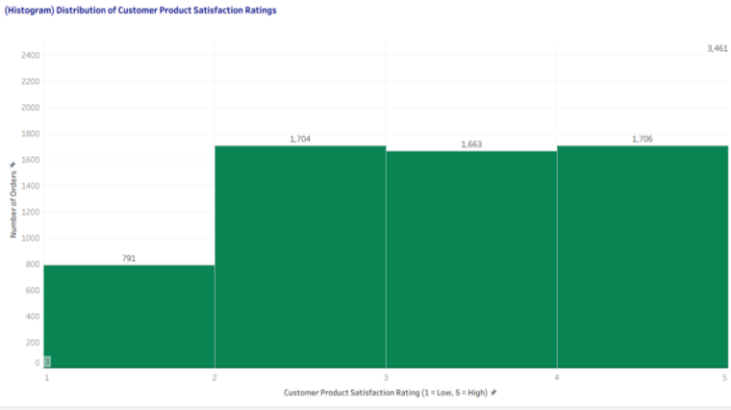


- The pie chart illustrates the proportion of total sales generated by new customers compared to returning customers. Each segment represents the share contributed by these two customer groups.
- This visualization highlights customer behavior patterns and helps understand whether the business relies more on acquiring new customers or retaining existing ones.
- A larger share from returning customers indicates customer loyalty and satisfaction with products or services.

Key Insight: Returning customers contribute a significant portion of total sales, suggesting that customer retention plays an important role in sustaining business revenue.

(Histogram) Distribution of Customer Product Satisfaction Ratings

[Tableau Public Link <click here>]

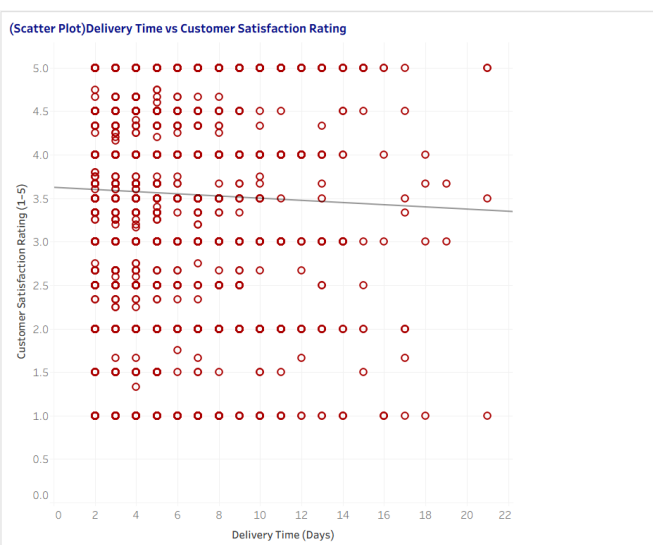


- The histogram shows how customer product satisfaction ratings are distributed across different rating levels. Each bar represents the frequency of ratings that fall within a specific satisfaction range.
- The distribution indicates that most ratings fall between 3 and 5, meaning the majority of customers report moderate to high satisfaction with the products.
- The pattern of the bars shows an increasing trend toward higher rating values, indicating that higher satisfaction scores occur more frequently than lower scores.
- The shape of the histogram suggests a left-skewed (negatively skewed) distribution, where the tail extends toward the lower rating values while most observations are concentrated at the higher end.

Key Insight: The histogram indicates that customer satisfaction is generally high, with ratings concentrated in the upper ranges. The slight negative skew suggests that fewer customers report low satisfaction, reflecting an overall positive customer experience with the products.

(SCATTER PLOT) DELIVERY TIME VS CUSTOMER SATISFACTION RATING

[Tableau Public Link <click here>]

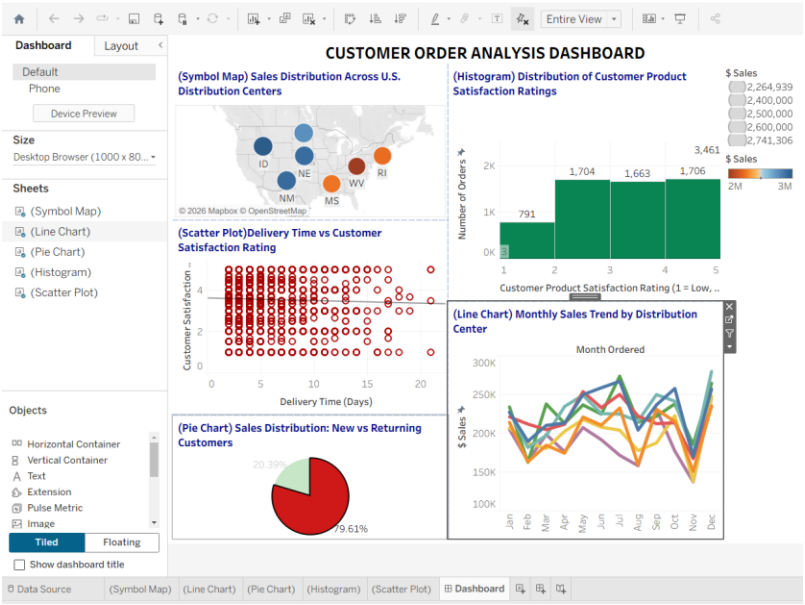


- The scatter plot analyzes the relationship between delivery time and customer satisfaction ratings. Each point represents an individual order, showing how satisfaction changes as delivery time increases.

- The distribution of points suggests a weak negative correlation between delivery time and customer satisfaction. As delivery time increases, satisfaction ratings tend to slightly decrease.
- The trend line in the visualization shows a slight downward trend, indicating that longer delivery durations are generally associated with somewhat lower satisfaction levels.
- Although the pattern is not extremely strong, the spread of points suggests that faster deliveries tend to maintain higher satisfaction scores, while delayed deliveries may reduce customer experience.

Key Insight: The analysis indicates that improving delivery speed can help maintain higher customer satisfaction levels, highlighting the importance of efficient logistics and timely order fulfillment.

TABLEAU DASHBOARD - [Tableau Dashboard Public Link <click here>]



Overall, the dashboard provides a comprehensive view of the business performance by combining multiple perspectives such as location-based sales activity, customer purchasing behavior, product satisfaction, and delivery efficiency. The visualizations help highlight key patterns that can assist in identifying areas of strong performance as well as opportunities for improvement in distribution and customer service operations.